ALGEBRA 2 · UNIT 4: POLYNOMIAL FUNCTIONS

Water Slide Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students analyze and design a water slide profile modeled by a cubic polynomial. They evaluate and graph it, connect zeros and multiplicity to a smooth landing, use synthetic division and the Remainder and Factor Theorems, apply the Rational Root Theorem, and design their own slide.

Multiplicity has a physical meaning here: a double zero makes the slide touch the water level without crossing it, which is a smooth landing. Synthetic division and the Remainder Theorem become quick ways to check heights.

Standards Addressed

- A-APR.2 Remainder Theorem
- A-APR.3 Identify zeros and use them to sketch a graph
- F-IF.7c Graph polynomial functions, showing end behavior
- A-SSE.2 Use structure to rewrite expressions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with slide photos. Part 1 table, graph, zeros.
Day 2	Part 2 division and theorems.
Day 3	Part 3 design.

Materials

- Student handout
- Graphing calculators or Desmos
- Graph paper

How the Three Versions Differ

Level	What changes
CP2	Evaluating, factored form, zeros and multiplicity, and synthetic division with a template.
CP1	Expanding, Remainder and Factor Theorems, the Rational Root Theorem, and end behavior.
Honors	A general design formula, a proof that the top is the highest point, and a design with constraints.

Watch For (Common Misconceptions)

- Missing a zero coefficient placeholder in synthetic division.
- Thinking a double zero crosses the axis.
- Reading end behavior from the wrong leading term.

Extension Ideas

- Measure a real slide from a photo.
- Add a hump using a degree-4 polynomial.
- Research ride safety rules.

Answer Key – CP2

Part 1: Read the Slide

#	Task	Answer
1	Complete the table.	32, 27, 16, 5, 0
2	How tall is the top of the slide? Where does the slide reach the pool?	32 ft high; reaches the pool at x = 40 ft
3	List the zeros of h and the multiplicity of each. Which zero is part of the slide?	x = 40 (multiplicity 2) and x = −20 (multiplicity 1). x = 40 is on the slide.
4	Explain why a double zero at x = 40 makes a smooth landing.	The graph touches the water level and turns instead of crossing, so the slide levels out.

Part 2: Check with Division

#	Task	Answer
5	Use synthetic division to divide $x^3 - 60x^2 + 0x + 32{,}000$ by $(x - 40)$. What is the remainder? What does it tell you?	Bottom row: 1, −20, −800, 0. Remainder 0, so $(x - 40)$ is a factor.
6	Factor the quotient to get all the factors.	$x^2 - 20x - 800 = (x - 40)(x + 20)$, so $h(x) = 0.001(x - 40)^2(x + 20)$
7	Use synthetic division to find the remainder when dividing by $(x - 10)$. Compare $0.001 \times$ remainder with h(10).	Bottom row: 1, −50, −500, 27,000. $0.001 \times 27{,}000 = 27 = h(10)$

Part 3: Design a Slide

#	Task	Answer
8	Use the form $h(x) = a(x - 30)^2(x + 15)$. Find a so that h(0) = 27.	$a(900)(15) = 27 \rightarrow a = 0.002$
9	Make a table for x = 0, 10, 20, 30.	27, $0.002(400)(25) = 20$, $0.002(100)(35) = 7$, 0
10	Sketch your slide and compare it with Splash Canyon's.	Answers vary.

Answer Key – CP1

Part 1: Read the Slide

#	Task	Answer
1	Expand to standard form. Check with x = 10.	$0.001x^3 - 0.06x^2 + 32$. h(10) = 1 − 6 + 32 = 27 ✓
2	Describe the end behavior of h (ignore the slide's domain).	As $x \to \infty$, $h \to \infty$; as $x \to -\infty$, $h \to -\infty$ (positive leading coefficient, odd degree).
3	Give the zeros, multiplicities, and the slide's domain and range.	Zeros 40 (double), −20 (single). Domain $0 \le x \le 40$; range $0 \le h \le 32$.
4	Rule: the slide must be at least 10 ft high for the first 25 ft. Does it follow the rule?	h(25) = 0.001(225)(45) ≈ 10.1 ft, and h decreases from 0 to 40, so yes.

Part 2: Theorems at Work

#	Task	Answer
5	Use the Remainder Theorem to find p(30) with synthetic division. Then find h(30).	Bottom row 1, −30, −900, 5,000: p(30) = 5,000; h(30) = 5 ft
6	List the possible rational roots of p according to the Rational Root Theorem (just describe the set). Then test x = −20.	Possible rational roots are ± the factors of 32,000. p(−20) = −8,000 − 24,000 + 32,000 = 0 ✓
7	Use the Factor Theorem and division to factor p completely.	$p(x) = (x + 20)(x^2 - 80x + 1600) = (x + 20)(x - 40)^2$

Part 3: Design a Slide

#	Task	Answer
8	Write $h(x) = a(x - 30)^2(x + 15)$ and find a.	a(900)(15) = 24 → a = 24 ÷ 13,500 = 2 ÷ 1125 ≈ 0.00178
9	Expand your polynomial to standard form.	Answers vary by rounding: $a(x^3 - 45x^2 + 13{,}500)$ with a ≈ 0.00178
10	Use synthetic division to verify a height at x = 10.	Answers vary.

Answer Key – Honors

Part 1: General Slide Formula

#	Task	Answer
1	Show that $h(0) = 0.5aL^3$ and find a in terms of H and L. Check with Splash Canyon (H = 32, L = 40).	$h(0) = a(L^2)(0.5L) = 0.5aL^3 = H \rightarrow a = 2H \div L^3$. Splash Canyon: $64 \div 64{,}000 = 0.001$ ✓
2	Expand h(x) and show $h(x) - H = a\,x^2(x - 1.5L)$.	$(x - L)^2(x + 0.5L) = x^3 - 1.5Lx^2 + 0.5L^3$, so $h(x) = ax^3 - 1.5aLx^2 + H$ and $h(x) - H = ax^2(x - 1.5L)$.
3	Use the identity to prove that the start (x = 0) is the highest point of the slide for $0 \le x \le L$.	For $0 \le x \le L$, $x^2 \ge 0$ and $x - 1.5L < 0$, so $h(x) - H \le 0$: $h(x) \le H$.

Part 2: Theorems

#	Task	Answer
4	Use the Rational Root Theorem and synthetic division to factor p completely.	$p(x) = (x + 20)(x - 40)^2$
5	Find the remainder when p is divided by (x − 30) and interpret it.	p(30) = 5,000, so h(30) = 5 ft.
6	Find the average rate of change of h from 0 to 20 and from 20 to 40. Where is the slide steeper?	0 to 20: $(16 - 32) \div 20 = -0.8$ ft per ft. 20 to 40: $(0 - 16) \div 20 = -0.8$. Same average steepness.

Part 3: Design Within Constraints

#	Task	Answer
7	Find the smallest L that satisfies the constraints (test values or solve).	$h(15) = (60 \div L^3)(15 - L)^2(15 + 0.5L)$. L = 25: ≈ 10.56 (too low); L = 26: ≈ 11.57 (too low); L = 27: ≈ 12.51 ✓. Smallest whole-number L is 27 ft.
8	Write the final polynomial and graph it.	Answers vary.
9	Write the report with your proofs.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown. Goes beyond the minimum with a creative, realistic choice.	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Polished, organized, and visually engaging.	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.